



STUDENT HANDBOOK

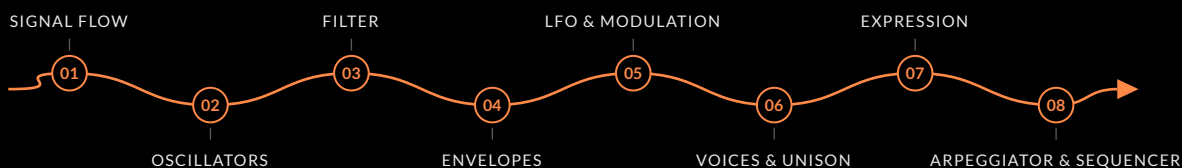
INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm. Everything from the workshop in one place: the theory, the diagrams, and the words.

A COMPANION HANDBOOK · WRITTEN BY IULIAN CAZAN

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SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL



THE COURSE MAP · EIGHT MODULES ALONG THE SIGNAL PATH



SINTEZAUR 8 MODULES · 90 MINUTES



SEQUENTIAL

BEFORE WE START

Two of you share one Fourm. One drives, one reads and listens; swap at every exercise, so both hands get to know the panel. Headphones on when you play; one ear off when the presenter talks. Set MAIN to about 12 o'clock and leave it there.

Every exercise starts from the same place: the Basic Program, a plain sawtooth with the filter open. If you get lost, come home to it: hold TRANSPOSE DOWN and press PROGRAM. Don't press PROGRAM otherwise — you would lose the sound you just built.

This handbook follows the workshop stop by stop. Each module has one page: the idea in a few lines, the diagram from the screen, and a TRY IT you can do at the synth. The words we use are collected in the glossary at the back. Read one module, turn one knob, listen.

SIGNAL FLOW

Every sound a subtractive synthesizer makes travels the same road: an oscillator generates a raw, harmonically rich tone, the mixer balances the sources, the filter sculpts the brightness, and the amplifier shapes the loudness over time. Master this path and every knob on the panel has an obvious home. Fourm is our instrument for the journey.

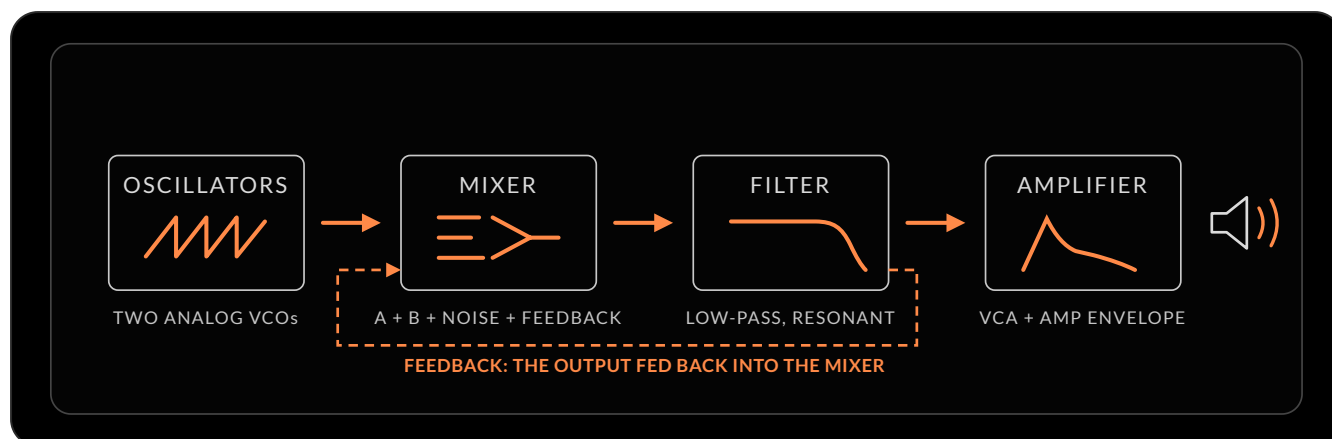


FIG. 1 · THE VOICE PATH: OSCILLATORS → MIXER → FILTER → AMPLIFIER → OUTPUT, WITH THE FEEDBACK RETURN.

WHAT'S IN A SOUND

A note is a fundamental frequency plus a stack of quieter overtones above it: the harmonics. Their mix is what you hear as timbre. A sawtooth carries every harmonic, a square only the odd ones, a triangle very few — which is why they sound bright, hollow and soft. Subtractive synthesis starts from a rich wave and takes harmonics away.

VCO: THE SOURCE

The voltage-controlled oscillator sets the pitch and the raw timbre. Fourm has two; brighter waveshapes carry more harmonics for the filter to work with.

VCF: THE SCULPTOR

The low-pass filter removes harmonics above the cutoff frequency, darkening the tone as you close it. Resonance emphasizes the region right at the cutoff. Its own module comes next: 03.

MIXER: FOUR SOURCES

Osc A, Osc B, the noise generator and the feedback return meet here. If nothing is turned up, nothing reaches the filter: every patch starts at the mixer.

VCA: THE HAND ON THE VOLUME

The voltage-controlled amplifier sets loudness moment by moment, driven by the amp envelope: silence before the key, a shaped rise and fall after it (module 04).

TRY IT From the Basic Program, hold a note and turn OSC A in the mixer all the way down: no source, no sound. Bring it back, then press SAW on Oscillator B and raise OSC B: two oscillators, one note, already thicker. With a finger, follow the road across the panel, left to right.

OSCILLATORS

Where every sound is born. Each of Fourm's two oscillators offers a set of waveshapes, selected with dedicated buttons and freely combined: Oscillator A has sawtooth and pulse, Oscillator B adds a triangle. The shape you choose decides the harmonic material everything downstream will carve into — and two humble additions, a sub layer and a pinch of noise, separate thin patches from full ones.

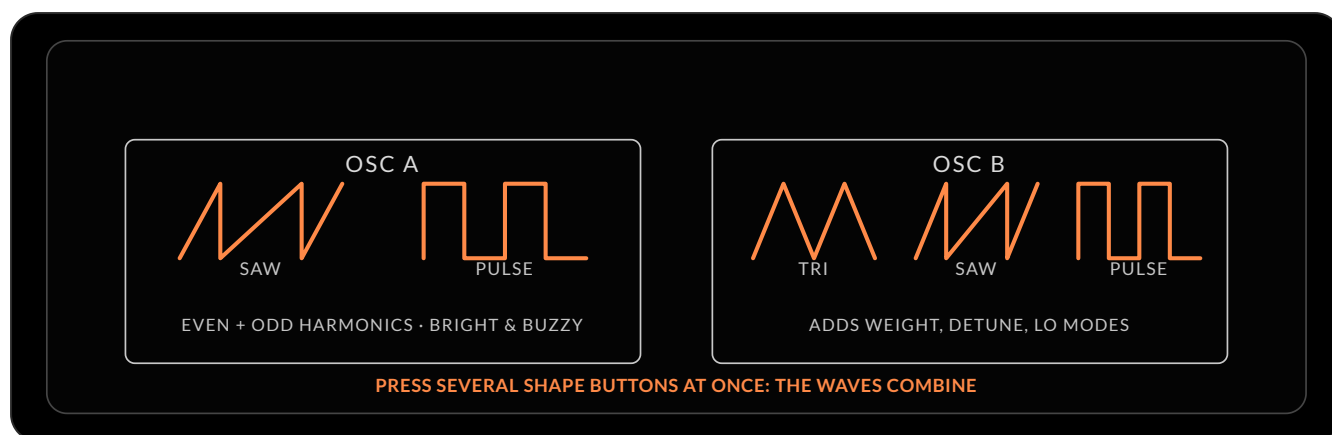


FIG. 2 • THE WAVESHAPES: SAWTOOTH, TRIANGLE, PULSE — AND THE SUB TRICK: OSC B, TRIANGLE, ONE OCTAVE DOWN.

SAW • TRI • PULSE

Sawtooth carries every harmonic: bright and full, the classic synth backbone, the sound of brass and strings. Triangle is soft and nearly pure, the natural sub layer. Pulse sits in between — hollow like a clarinet at square, thin and reedy as it narrows — and its character depends on its width.

PULSE WIDTH & PWM

Pulse width is the duty cycle of the pulse wave, from square to a thin sliver. Fourm has no dedicated knob: hold PULSE and turn SELECT (0 is square; toward ± 100 the pulse thins until it disappears). Modulating the width with the LFO — pulse width modulation — keeps the tone in constant, chorus-like motion from a single oscillator. We do it in module 05.

LEVEL 2

Below the octaves, Osc B has two low modes: LO1 still follows the keyboard, LO2 free-runs. Down there it stops being a voice and becomes a modulation source, like an extra, polyphonic LFO. Sync locks Oscillator A's cycle to Oscillator B for aggressive, tearing timbres.

TRY IT From the Basic Program: on Oscillator A press PULSE so both saw and pulse are lit, then hold PULSE and turn SELECT toward the edge and back. On Oscillator B press TRI, set the octave to -1 and raise OSC B in the mixer: add it, take it away, feel the weight go. Nudge Osc B's FREQUENCY slightly off centre and listen to the beating; too far and it splits into two notes. Press NOISE, hold it and turn SELECT down to about 30 — just a breath of air.

OCTAVES & TUNING

Each oscillator has five octave positions, from -2 to +2. Setting A and B an octave apart is the fastest way to a bigger sound. FREQUENCY fine-tunes by up to seven semitones either way; a few cents of detune between A and B makes the two waves beat against each other, and the tone starts to breathe. Push it too far and you hear two notes — bring it back until it just shimmers.

THE SUB TRICK, AND NOISE

Set Osc B to triangle, one octave below Osc A: the nearly pure wave slips under the main tone and adds weight without clutter. Noise is the only source with no pitch at all — white (full band), pink (warmer), violet (tilted to the highs); press NOISE, then hold it and turn SELECT for level. A trace sharpens attacks and adds breath; more of it becomes percussion.

FILTER

Shaping your tone. The oscillators give you a block of harmonics; the filter is the chisel. Fourm's low-pass filter lets everything below a chosen frequency through and fades everything above it — close it and the sound darkens, open it and it brightens. It is the most important knob on any synthesizer, and the one you will touch most tonight.

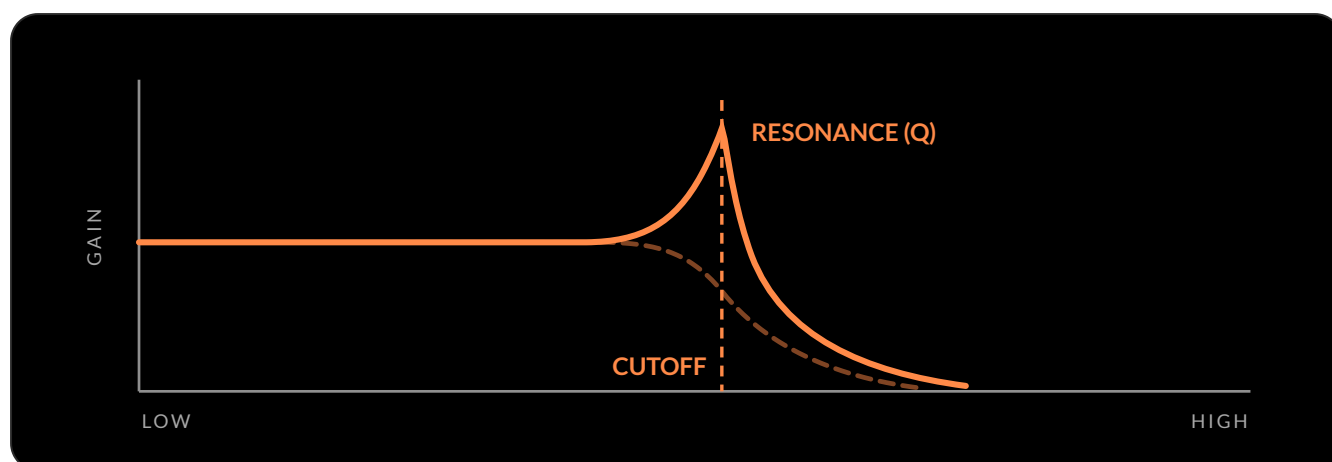


FIG. 3 · A LOW-PASS FILTER WITHOUT AND WITH RESONANCE: THE CUTOFF, AND THE PEAK THAT SINGS.

CUTOFF

The hinge frequency. Below it the harmonics pass; above it they fall away, on Fourm with a steep four-pole slope. Fully open, you hear the raw wave; fully closed, you hear nothing at all — the first thing to check when a patch goes silent.

SELF-OSCILLATION

At maximum resonance the filter feeds on itself and produces a pure sine of its own, with no oscillator needed: a fifth sound source. Turn the volume down a touch before you go there — it arrives suddenly and loud.

RESONANCE

A boost of the narrow band right at the cutoff. A little adds focus and vowel-like character to a sweep; a lot makes the filter whistle along with the note. Unlike many classic filters, Fourm keeps the low end full as resonance rises.

KEY TRACK & ENV AMOUNT

KEY TRACK makes the cutoff follow the keyboard (quarter, half or full — hold the button and turn SELECT): higher notes get a brighter filter, as acoustic instruments do; with full tracking, a singing filter plays a scale in tune. ENV AMT sets how much the filter envelope moves the cutoff over the life of a note — positive or negative, with zero at 12 o'clock. That is the story of the next module.

LEVEL 2

Fourm's filter is a four-pole resonant low-pass in the Prophet-5 lineage. Its bass compensation is always on, so resonance never thins the fundamental the way vintage ladder filters do.

TRY IT Hold a chord. Turn CUTOFF slowly down until the sound is dark, then back up. Set RESONANCE to about 3 o'clock and sweep CUTOFF again: a whistle rides the sweep. Lower MAIN a touch, turn RESONANCE to maximum, close CUTOFF a little and play single notes: the filter sings on its own. Hold KEY TRACK, turn SELECT to Full, and hear the high notes brighten. Bring RESONANCE back to 9 o'clock and CUTOFF to 12 o'clock — we keep this for the next module.

ENVELOPES

Controlling your sound over time. An oscillator through a filter is still a beep: it starts when you press and stops when you let go, and nothing happens in between. Real instruments have a shape — an organ starts and stops instantly, a plucked string starts loud and fades, a bowed note swells. The envelope is that shape, drawn with four knobs and replayed every time you press a key.

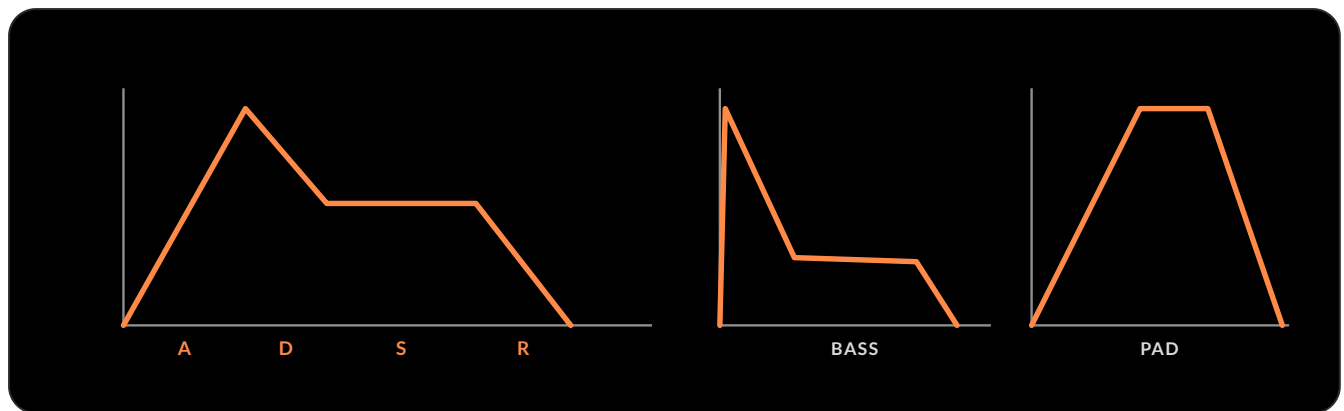


FIG. 4 • THE ADSR ENVELOPE, AND TWO SHAPES: BASS AND PAD.

ADSR

Attack is the time to rise from silence to the peak. Decay is the time to fall from the peak to the sustain level. Sustain is a level, not a time: where the sound sits for as long as the key is held. Release is the time to fade to silence after the key is let go. Your brain identifies an instrument in its first milliseconds — attack matters most.

THE COOKBOOK

Sustain at maximum: the decay has nowhere to fall and does nothing. Sustain at zero: the note dies while you hold it — a pluck. Attack and sustain at zero: percussion. Filter envelope slower than amp envelope: a pad that opens up; the other way round you hear nothing of it, because the amp closes first. Filter decay is the pluck knob: shorten it and the sound bites.

TRY IT

RECIPE 1 • BASS

Start from the module-02 sound (A saw + pulse, B triangle an octave down). Amp envelope: ATTACK 0, DECAY 10 o'clock, SUSTAIN 8 o'clock, RELEASE 8 o'clock. Filter envelope: ATTACK 0, DECAY 10 o'clock, SUSTAIN minimum, RELEASE 8 o'clock; in the filter, ENV AMT to 3 o'clock and CUTOFF to 9 o'clock. Play short, low notes. Now shorten the filter DECAY: more pluck.

TWO ENVELOPES ON FOURM

The AMP ENVELOPE drives the amplifier: the loudness contour of a note. The FILTER ENVELOPE drives the cutoff, scaled by ENV AMT in the filter section: the brightness contour. Same shape, two jobs. ENV AMT is bipolar — zero at 12 o'clock, positive clockwise, inverted counter-clockwise; an amount left at 12 o'clock is the classic “why is nothing happening”.

VELOCITY

Each envelope has a VELOCITY button: how hard you strike scales that envelope's amount (hold it and turn SELECT for the depth; default 127). Velocity to the filter envelope is why soft playing stays dark and hard playing bites — module 07.

RECIPE 2 • PAD

Amp envelope: ATTACK 12 o'clock, DECAY 0, SUSTAIN maximum, RELEASE 2 o'clock. CUTOFF to 12 o'clock, ENV AMT to 10 o'clock; Osc B to saw at octave 0, FREQUENCY nudged slightly for detune. Hold a chord and let it bloom. (Knob positions are starting points, not rules — trust your ears.)

LFO & MODULATION

Adding movement. Envelopes give a note a shape; modulation keeps it alive. A low frequency oscillator is an oscillator too slow to hear as a pitch — you hear what it moves. Send it to the pitch and you get vibrato, the violinist rocking a finger on the string; to the cutoff, a wah; to the amplifier, tremolo, the volume knob swung twice a second. Modulation is the verb of synthesis: something moves something else.

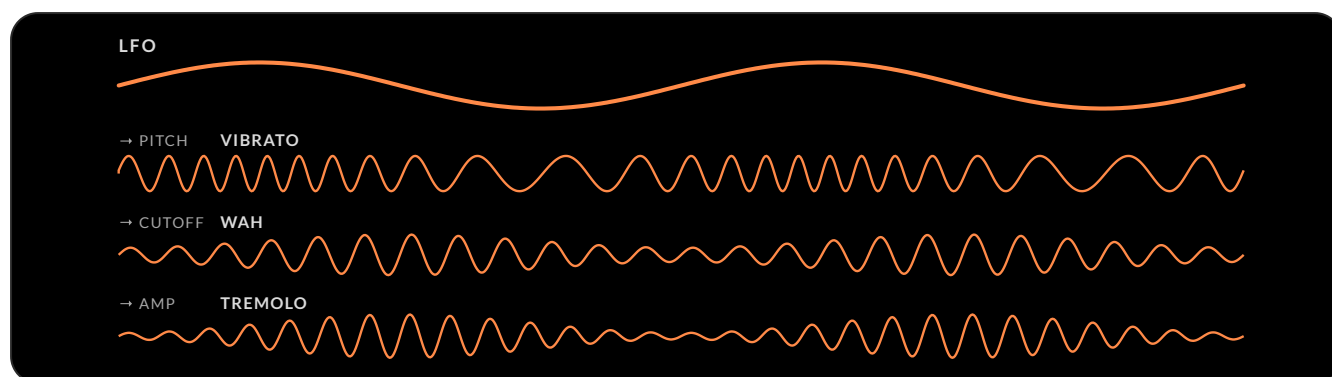


FIG. 5 · ONE LFO, THREE DESTINATIONS: VIBRATO, WAH, TREMOLO.

THE LFO

Its FREQUENCY knob is the speed, from a slow drift to a fast buzz; SYNC locks it to the clock. Its shapes stack: triangle for smooth vibrato and tremolo, sawtooth for ramps, square for trills and on-off tremolo, sample & hold for stepped random motion. Musical vibrato is small and around six to seven cycles per second; speed it up and it stops being vibrato and becomes a growl.

THE CLASSIC MOVES

Vibrato: LFO triangle → FREQ A and FREQ B, a small amount, on the wheel. Wah: LFO → CUTOFF, a sizable amount, slow. Tremolo: LFO square or triangle → AMP. PWM: LFO → PW A, the pulse that never stands still.

LEVEL 2

Osc B in LO1 or LO2 is a second, polyphonic LFO — each voice gets its own rate. Osc B modulates even with its mixer level at zero. The filter envelope is a modulation source too: FILT ENV → FREQ A gives the “pitch blip” of synth brass; → PW A, a pulse that narrows over the note.

TRY IT On the pad from module 04: in the LFO section press TRIANGLE and set FREQUENCY to about 10 o'clock. In the modulation section press ROUTE under LFO twice (blue), then FREQ A and FREQ B twice each (blue). Turn the LFO AMOUNT from 12 o'clock clockwise until you hear it, then push the mod wheel: vibrato that you control. Now press FREQ A and FREQ B until they go dark, press PW A twice — the pulse breathes; then CUTOFF twice — wah. Faster FREQUENCY, more AMOUNT: go too far, then bring it back.

THE MODULATION SECTION

Three sources on the left — FILT ENV, OSC B and the LFO — each with a ROUTE button and a bipolar AMOUNT knob (zero at 12 o'clock). Eight destinations on the right: FREQ A, FREQ B, PW A, PW B, CUTOFF, AMP, LFO FREQ, LFO AMT. Press ROUTE once for the direct bus (LED red) and the modulation is always on; press it twice for the mod-wheel bus (blue), and it only appears as you raise the wheel — the wheel becomes the volume of the modulation. Destinations work the same way: once for the direct bus, twice for the wheel bus, three times for both (purple).

VOICES & UNISON

Four voices, one note, and the loop that thickens everything. Fourm is polyphonic: it holds four complete voices — each with its own oscillators, filter and envelopes — so four notes can sound at once. This module is about how those voices are spent: spread across a chord, stacked on one massive note, or replayed as a stored chord from a single key. And it adds mass from a different place: the feedback loop.

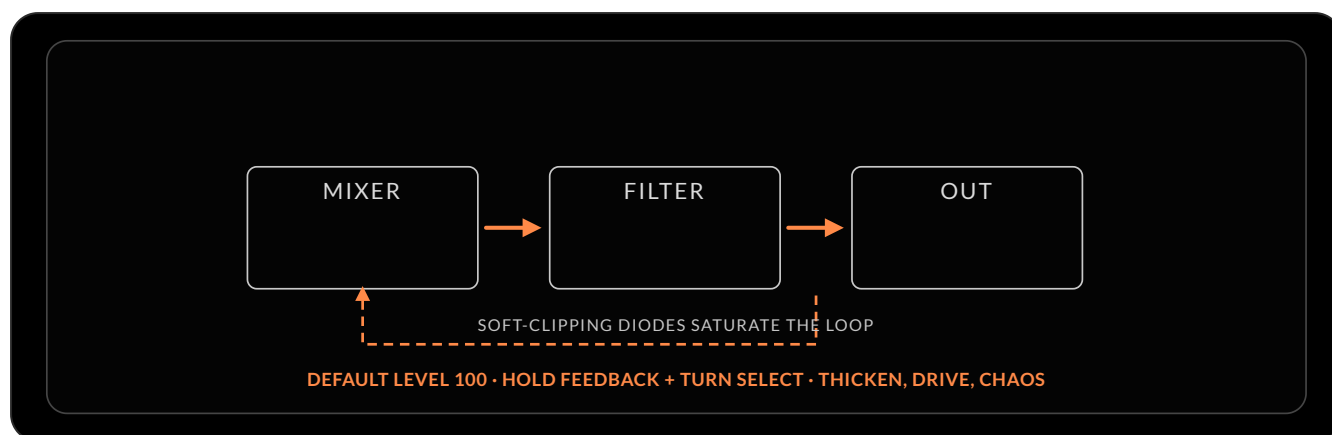


FIG. 6 · FEEDBACK: THE OUTPUT RETURNED INTO THE MIXER THROUGH SOFT CLIPPING.

POLYPHONY

Mono synths play one note at a time. Fourm plays four; press a fifth and the oldest note is stolen — faster than you expect once unison is on. HOLD sustains up to four held notes after your hands leave the keys.

CHORD MEMORY

Hold a chord and store it; from then on any single key replays the whole voicing, transposed as you move — one finger, full harmony, great for chord stabs. It is saved with the program.

UNISON

Unison stacks the voices you choose — one to four (hold UNISON and turn SELECT) — on a single note. Fourm becomes monophonic and, with all four, enormous. On its own that is just louder; the magic is detune (Param 10, Unison Detune, 0–127), which spreads the voices a few cents apart into one wide, beating tone. Small reads as analog richness, large as an obvious chorus. When it sounds huge, back it off ten percent: that is usually the keeper.

FEEDBACK

A dedicated mixer channel returns Fourm's output into the mix, with soft-clipping diodes in the loop. Press FEEDBACK (it comes in at level 100), hold it and turn SELECT to set the amount. Low amounts warm and thicken; high amounts growl and overdrive, all inside the voice. Unison plus feedback is Fourm's biggest bass.

LEVEL 2

Chord memory, step by step: press UNISON; push SELECT to move to the middle row and scroll to Chord; press UNISON to move to the top row; play and hold a chord of two to four notes; press UNISON again — Chord Mode is on. To clear it: hold UNISON, set the voice count with SELECT, press UNISON off, save. Vintage (Param 11) adds voice-to-voice drift without unison, like an old synth on a warm night.

TRY IT Play five notes at once and count what you hear. Hold UNISON, turn SELECT to 4 voices and play a line: one huge note. Press PARAM, scroll to 10 Unison Detune, press SELECT and turn it to about 40, press PROGRAM to exit — the tone widens. Press FEEDBACK, hold it and turn SELECT down to about 60: growl without chaos. Turn unison off, play a chord, press HOLD and lift your hands.

EXPRESSION

Play it, don't just press it. Envelopes and LFOs make every note the same; what makes it music is what your hands add. Velocity reads how fast you strike, aftertouch reads how hard you keep pressing, glide lets one note slide into the next. Fourm's Tactive keybed senses pressure under every single key — polyphonic aftertouch — so each finger shapes its own note.

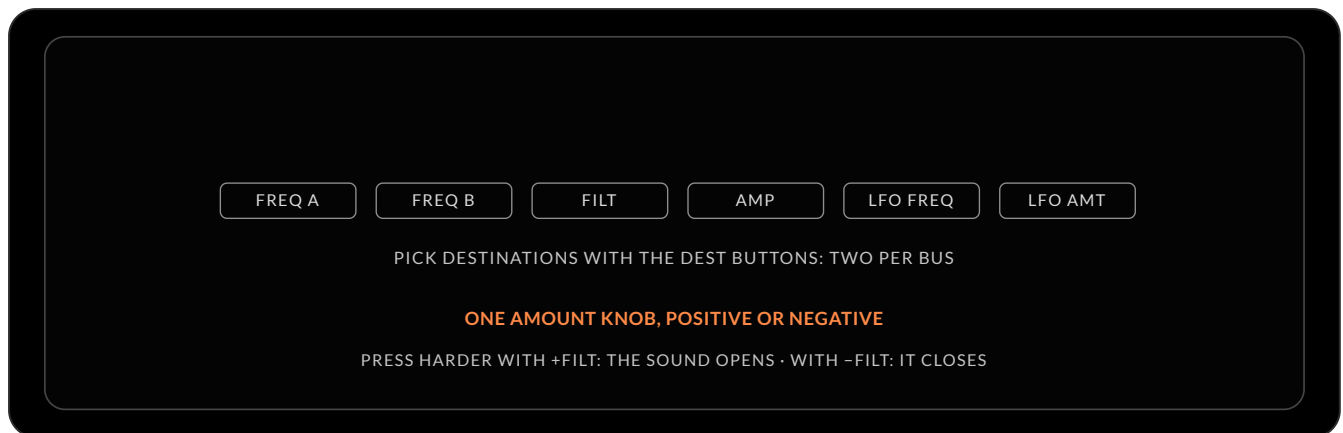


FIG. 7 • PRESSURE ROUTING: ONE AMOUNT, SIX DESTINATIONS.

VELOCITY

Louder notes on real instruments also have more harmonics, which is why velocity belongs on the filter, not just the volume. Each envelope has a VELOCITY button: press it, and how hard you play scales that envelope's depth (hold it and turn SELECT for the amount). Velocity to the filter envelope: soft playing stays dark, hard playing bites. Five response curves in Global — Soft+, Soft, Medium, Hard, Hard+ — match the keybed to your touch.

PRESSURE ROUTING

Six destinations — FREQ A, FREQ B, FILT, AMP, LFO FREQ, LFO AMT — chosen with the three DEST buttons, two per bus, and one AMOUNT knob, positive or negative. Cutoff is the classic: press for brightness, release to relax; a negative amount closes the filter instead. Pitch is for the brave.

POLY AFTERTOUCH

After the key is down, extra pressure becomes a continuous modulator, per key. Lean into one note of a held chord and only that note blooms — like blowing harder into a flute: press harder, brighter. Mono aftertouch, on most keyboards, moves every voice at once; per-note aftertouch used to be a rarity — the CS-80, the Prophet T8 — and Fourm has it natively.

GLIDE

Also called portamento: the pitch slides from the previous note to the next instead of jumping. Press GLIDE and set RATE — glide on with rate at zero does nothing. Subtle settings, barely a slur, sound the most musical: it is the difference between a keyboard melody and a phrase that sounds like a voice.

LEVEL 2

Four glide modes (hold GLIDE and turn SELECT): Fixed Rate (default; larger intervals take longer), Fixed Time (any interval takes the same time), and their legato variants (glide only when notes overlap). The legato and fixed-time modes appear when Unison is on. Aftertouch curves live in Global too.

TRY IT In the FILTER ENVELOPE press VELOCITY; play soft, then hard. In AFTERTOUCH press the DEST button under FILT / AMP until only FILT is lit, turn AMOUNT to 2 o'clock; hold a three-note chord and press into just the top note. Now turn AMOUNT to 10 o'clock: pressure closes the filter instead. Press GLIDE, set RATE to 10 o'clock and play a bass line: the notes slide. Then move the pressure to FREQ A and FREQ B with a small amount — a bend, per finger.

ARPEGGIATOR & SEQUENCER

Patterns from held notes. Hold a chord and the arpeggiator plays it back one note at a time, locked to the clock. It turns sustained harmony into rhythm — and Fourm hides a full sequencer behind it. The mouse is not a musical instrument; this is why the arpeggiator lives on the panel.

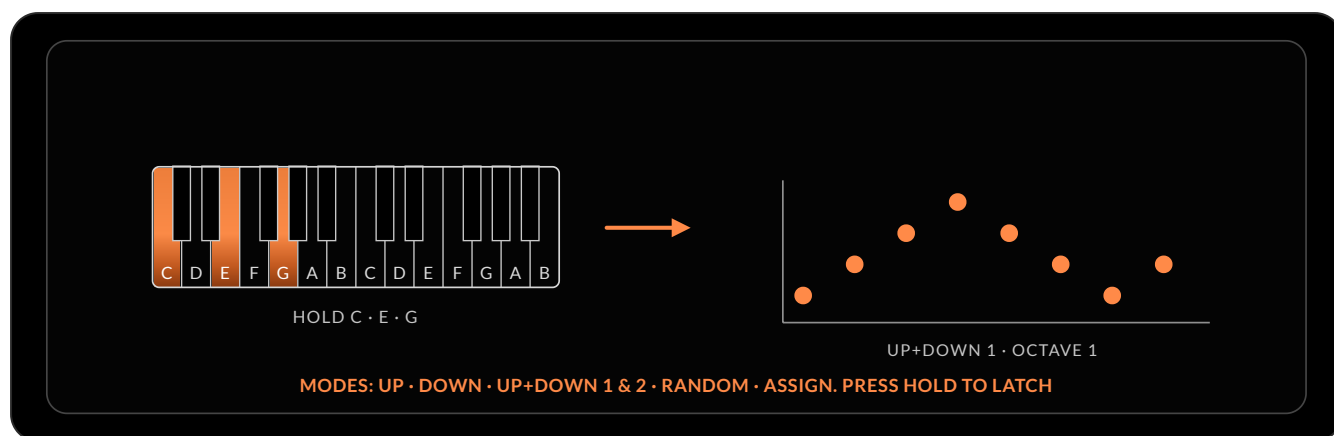


FIG. 8 • HOLD A CHORD, THE SYNTH PLAYS THE PATTERN.

DIRECTIONS & OCTAVES

Press ON in the arpeggiator and hold some notes. Six modes — Up, Down, Up+Down 1 and 2, Random, Assign (as played) — over one to three octaves (hold MODE or OCTAVE and turn SELECT). HOLD keeps the pattern running while both hands leave the keys; play a different number of notes and the rhythm changes — it sounds like a mistake until you embrace it.

LIVE MOVES

While a pattern runs, sweep the cutoff, lean into aftertouch, flip divisions. The arpeggiator supplies the rhythm; you play the tone.

CLOCK & DIVISIONS

One clock rules the tempo: tap TAP TEMPO four times, or hold it and turn SELECT for a BPM. CLK DIV (hold it and turn SELECT) picks the rhythmic grid — quarters, eighths, sixteenths, dotted, triplets — and the LFO's SYNC follows the same clock. If the arp will not start, check Clock Mode in Global: set to In, it waits for an external clock.

NOTE & MOD SEQ (LEVEL 2)

Beyond arpeggios, Fourm records up to 64 steps of notes — chords, rests, ties, even a glide per step — triggered and transposed from the keyboard, and a mod sequence can step a parameter instead: cutoff, pulse width, feedback. Each sequencer track is tied to a voice; put chords on a step for polyphony. A sequencer in disguise, and a story for another workshop.

TRY IT Press ON in the arpeggiator, hold C–E–G, press HOLD and lift your hands. Tap TAP TEMPO four times to your tempo. Change CLK DIV to sixteenths (hold it and turn SELECT), then back to eighths. Ride CUTOFF while it runs. Hold MODE and turn SELECT to Random; hold OCTAVE and turn SELECT to 2. Now hold a “wrong” chord and let it repeat: give it eight bars.

FUNDAMENTALS

SIGNAL FLOW

The road a sound takes: oscillators → mixer → filter → amplifier → output.

VCO • OSCILLATOR

Voltage-controlled oscillator: generates the raw waveform. Fourm has two.

SQUARE

Pulse at 50% width: odd harmonics only, hollow and woody.

PULSE WIDTH • PWM

The duty cycle of the pulse wave (hold PULSE, turn SELECT); PWM is modulating it.

MIXER

Sets the level of each source: Osc A, Osc B, noise, feedback.

SUB OSCILLATOR

A layer one octave below the main pitch; on Fourm, Osc B triangle does the job.

RESONANCE

A boost of the band right at the cutoff; high settings ring and can self-oscillate.

KEY TRACK

The cutoff follows the keyboard: off, quarter, half or full.

ENVELOPE • ADSR

Attack, Decay, Sustain, Release. Sustain is a level, not a time.

HARMONICS

Quieter overtones above the fundamental; their mix is what you hear as timbre.

SAWTOOTH

Every harmonic, falling at 1/n: bright and full, the classic synth backbone.

TRIANGLE

Odd harmonics that fade fast: soft, nearly a sine, the natural sub layer.

OCTAVE

Five positions per oscillator, -2 to +2; FREQUENCY fine-tunes up to seven semitones.

NOISE

Unpitched sound in white, pink or violet: air, breath and percussion.

VCF • CUTOFF

Voltage-controlled filter; cutoff is where the low-pass starts removing highs.

SELF-OSCILLATION

At maximum resonance the filter sings a pure sine of its own.

ENV AMOUNT

How much the filter envelope moves the cutoff; bipolar, zero at 12 o'clock.

VCA

Voltage-controlled amplifier: the loudness stage, shaped by the amp envelope.

PERFORMANCE & MOTION

LFO

A low frequency oscillator, too slow to hear as pitch: you hear what it moves.

VIBRATO

LFO to pitch.

MODULATION BUS

Direct (red LED) is always on; mod wheel (blue) appears as you raise the wheel.

POLYPHONY • VOICE

Fourm holds four complete voices; a fifth note steals the oldest.

DETUNE

Small pitch offsets between stacked voices, measured in cents; beating equals width.

HOLD

Sustains up to four notes after your hands leave the keys; also latches the arpeggiator.

VELOCITY

How fast the key goes down; five curves, from Soft+ to Hard+.

PRESSURE ROUTING

Six destinations on three DEST buttons, with one bipolar AMOUNT knob.

LEGATO

Overlapping notes; in the legato glide modes, only these slide.

NOTE & MOD SEQ

Up to 64 recorded steps of notes, or a stepped lane that moves a parameter.

RATE

The LFO's speed; SYNC locks it to the clock.

TREMOLO

LFO to amplitude.

MOD WHEEL

The volume of the modulation on the wheel bus.

UNISON

One to four voices stacked on a single note; the synth becomes monophonic.

CHORD MEMORY

A stored chord replayed from a single key, transposed as you move.

FEEDBACK

The output fed back into the mixer through soft-clipping diodes: warmth, then growl.

POLY AFTERTOUCH

Pressure sensed per key by the Tactive keybed; each finger modulates its own note.

GLIDE • PORTAMENTO

The pitch slides between notes instead of jumping; four modes.

CLOCK • BPM • DIVISION

The shared tempo and the rhythmic grid the arpeggiator and sequences follow.

TAKE IT HOME

That is the whole course. Keep this handbook next to the synth: read one module, turn one knob, listen. The order on the panel is the order in your head now — and every synthesizer you meet from here on is the same road with different clothes.

When you build a sound you like, write it down: the waveshapes, the octaves, the filter, the four knobs of each envelope, what the LFO moves. A patch on paper survives; a patch in memory rarely does.

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SINTEZAURthe community, the next workshop, and where to say hello [\[LINK\]](#)

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ZEEDOwhere to try and buy the Fourm and everything around it [\[LINK\]](#)

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SEQUENTIALthe Fourm user guide, the preset list and the instruments that made tonight possible [\[LINK\]](#)

Read next: Synth Secrets (Sound On Sound, free online) and Rick Snoman's Dance Music Manual — everything from tonight, in depth.

Thank you for playing.

